Enrollment No.: 240840116065

Practical-9

Double-click (or enter) to edit

```
# STEP 1: INSTALL AND IMPORT LIBRARIES

!pip -q install pandas plotly dash openpyxl ipywidgets
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go

from IPython.display import display, HTML
from ipywidgets import interact, Dropdown
from google.colab import files

print("Libraries imported successfully!")


# STEP 2: LOAD EXCEL DATASET

print("\nUpload your Superstore Excel Dataset")

uploaded = files.upload()

file_name = list(uploaded.keys())[0]

df = pd.read_excel(file_name)

print("\nFirst five rows of the dataset:")
display(df.head())

print("Dataset shape:", df.shape)
print("Column names:", df.columns.tolist())


# STEP 3: CLEAN AND PREPROCESS DATA

print("\nMissing values before cleaning:")
print(df.isnull().sum())

df = df.drop_duplicates().copy()

df["Order Date"] = pd.to_datetime(
    df["Order Date"], errors="coerce"
)
```

```
)

df["Ship Date"] = pd.to_datetime(
    df["Ship Date"], errors="coerce"
)

df = df.dropna(
    subset=["Order Date", "Sales", "Profit", "Category"]
)

print("\nCleaned Dataset:")
display(df.head())

print("\nDataset Information:")
df.info()


# STEP 4: EXPLORATORY DATA ANALYSIS (EDA)

total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
total_orders = df["Order ID"].nunique()

print("\nBUSINESS PERFORMANCE SUMMARY")
print("Total Sales:", round(total_sales, 2))
print("Total Profit:", round(total_profit, 2))
print("Total Unique Orders:", total_orders)

print("\nSTATISTICAL SUMMARY")

display(
    df[["Sales", "Profit", "Quantity", "Discount"]].describe()
)

category_sales = df.groupby("Category")["Sales"].sum()

print("\nSALES BY CATEGORY")
display(category_sales)


# STEP 5: INTERACTIVE VISUALIZATIONS

# 5.1 Sales by Category

category_data = df.groupby(
    "Category", as_index=False
)["Sales"].sum()

fig1 = px.bar(
    category_data,
    x="Category",
    y="Sales",
```

```
    color="Category",
    title="Total Sales by Category"
)

fig1.show()


# 5.2 Monthly Sales Trend

monthly_sales = (
    df.set_index("Order Date")
    .resample("MS")["Sales"]
    .sum()
    .reset_index()
)

fig2 = px.line(
    monthly_sales,
    x="Order Date",
    y="Sales",
    markers=True,
    title="Monthly Sales Trend"
)

fig2.show()


# 5.3 Profit by Region

region_profit = df.groupby(
    "Region", as_index=False
)["Profit"].sum()

fig3 = px.bar(
    region_profit,
    x="Region",
    y="Profit",
    color="Region",
    title="Profit by Region"
)

fig3.show()


# 5.4 Sales vs Profit

fig4 = px.scatter(
    df,
    x="Sales",
    y="Profit",
    color="Category",
```

```
    hover_data=["Region", "Sub-Category"],
    title="Sales vs Profit"
)

fig4.show()


# STEP 6: INTERACTIVE BUSINESS DASHBOARD

def create_dashboard(category, region):

    filtered = df.copy()

    if category != "All":
        filtered = filtered[
            filtered["Category"] == category
        ]

    if region != "All":
        filtered = filtered[
            filtered["Region"] == region
        ]

    total_sales = filtered["Sales"].sum()
    total_profit = filtered["Profit"].sum()
    total_orders = filtered["Order ID"].nunique()

    display(HTML(f"""
    <h1 style="text-align:center;color:#2563eb;">
        SUPERSTORE SALES DASHBOARD
    </h1>

    <h3>Total Sales: ${total_sales:,.2f}</h3>
    <h3>Total Profit: ${total_profit:,.2f}</h3>
    <h3>Total Orders: {total_orders:,}</h3>
    """))

    if filtered.empty:
        print("No data available for selected filters.")
        return

    # Sales by Category

    category_data = filtered.groupby(
        "Category", as_index=False
    )["Sales"].sum()

    display(px.bar(
        category_data,
        x="Category",
        y="Sales",
        color="Category"
```

```
        color="Category",
        title="Sales by Category"
    ))

    # Sales by Region

    region_data = filtered.groupby(
        "Region", as_index=False
    )["Sales"].sum()

    display(px.bar(
        region_data,
        x="Region",
        y="Sales",
        color="Region",
        title="Sales by Region"
    ))

    # Monthly Sales Trend

    monthly_data = (
        filtered.set_index("Order Date")
        .resample("MS")["Sales"]
        .sum()
        .reset_index()
    )

    display(px.line(
        monthly_data,
        x="Order Date",
        y="Sales",
        markers=True,
        title="Monthly Sales Trend"
    ))

    # Profit by Sub-Category

    profit_data = filtered.groupby(
        "Sub-Category", as_index=False
    )["Profit"].sum().sort_values("Profit")

    display(px.bar(
        profit_data,
        x="Profit",
        y="Sub-Category",
        orientation="h",
        color="Profit",
        title="Profit by Sub-Category"
    ))


# Interactive Filters
```

```
interact(
    create_dashboard,

    category=Dropdown(
        options=["All"] + sorted(
            df["Category"].dropna().unique().tolist()
        ),
        value="All",
        description="Category:"
    ),

    region=Dropdown(
        options=["All"] + sorted(
            df["Region"].dropna().unique().tolist()
        ),
        value="All",
        description="Region:"
    )
)


# STEP 7: ACTIONABLE INSIGHTS

best_category = (
    df.groupby("Category")["Sales"].sum().idxmax()
)

best_region = (
    df.groupby("Region")["Profit"].sum().idxmax()
)

lowest_profit_subcategory = (
    df.groupby("Sub-Category")["Profit"].sum().idxmin()
)

print("\nACTIONABLE INSIGHTS")

print("1. Highest sales category:", best_category)

print("2. Highest profit region:", best_region)

print(
    "3. Sub-category needing profit investigation:",
    lowest_profit_subcategory
)


# ETHICAL CONSIDERATIONS

print("\nETHICAL CONSIDERATIONS")
```

```
print("1. Protect customer privacy and confidential information.")
print("2. Use accurate data and document cleaning steps.")
print("3. Avoid unfair decisions based on incomplete or biased data.")
print("4. Interpret charts carefully and do not assume correlation proves causation.")
print("5. Explain dataset limitations when presenting results.")

# FINAL RESULT
print("\nPRACTICAL 9 COMPLETED SUCCESSFULLY!")
print("Interactive Business Dashboard created successfully.")
print("Total Sales:", round(df["Sales"].sum(), 2))
print("Total Profit:", round(df["Profit"].sum(), 2))
print("Total Orders:", df["Order ID"].nunique())
```

```
Libraries imported successfully!

Upload your Superstore Excel Dataset
```
Choose Files sample_-_superstore.xls

sample_-_superstore.xls(application/vnd.ms-excel) - 3422208 bytes, last modified: 10/31/2026 - 100% done

```
Saving sample_-_superstore.xls to sample_-_superstore.xls

First five rows of the dataset:
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	...	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Qu
0	1	US-2023-103800	2023-01-03	2023-01-07	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston	...	77095	Central	OFF-PA-10000174	Office Supplies	Paper	Message Book, Wirebound, Four 5 1/2" X 4" Form...	16.448	
1	2	US-2023-112326	2023-01-04	2023-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...	60540	Central	OFF-BI-10004094	Office Supplies	Binders	GBC Standard Plastic Binding Systems Combs	3.540	
2	3	US-2023-112326	2023-01-04	2023-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...	60540	Central	OFF-LA-10003223	Office Supplies	Labels	Avery 508	11.784	
3	4	US-2023-112326	2023-01-04	2023-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...	60540	Central	OFF-ST-10002743	Office Supplies	Storage	SAFCO Boltless Steel Shelving	272.736	
4	5	US-2023-141817	2023-01-05	2023-01-12	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelphia	...	19143	East	OFF-AR-10003478	Office Supplies	Art	Avery Hi-Liter EverBold Pen Style Fluorescent ...	19.536	

5 rows × 21 columns

```
Dataset shape: (10194, 21)
Column names: ['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country/Region', 'City', 'State/Province', 'Pos

Missing values before cleaning:
Row ID            0
Order ID          0
Order Date        0
Ship Date         0
Ship Mode         0
Customer ID       0
Customer Name     0
Segment           0
Country/Region    0
City              0
```

```
State/Province    0
Postal Code       0
Region            0
Product ID        0
Category          0
Sub-Category      0
Product Name      0
Sales             0
Quantity          0
Discount          0
Profit            0
dtype: int64

Cleaned Dataset:
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	...	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Qu
0	1	US-2023-103800	2023-01-03	2023-01-07	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston	...	77095	Central	OFF-PA-10000174	Office Supplies	Paper	Message Book, Wirebound, Four 5 1/2" X 4" Form...	16.448	
1	2	US-2023-112326	2023-01-04	2023-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...	60540	Central	OFF-BI-10004094	Office Supplies	Binders	GBC Standard Plastic Binding Systems Combs	3.540	
2	3	US-2023-112326	2023-01-04	2023-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...	60540	Central	OFF-LA-10003223	Office Supplies	Labels	Avery 508	11.784	
3	4	US-2023-112326	2023-01-04	2023-01-08	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...	60540	Central	OFF-ST-10002743	Office Supplies	Storage	SAFCO Boltless Steel Shelving	272.736	
4	5	US-2023-141817	2023-01-05	2023-01-12	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelphia	...	19143	East	OFF-AR-10003478	Office Supplies	Art	Avery Hi-Liter EverBold Pen Style Fluorescent ...	19.536	

5 rows × 21 columns

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10194 entries, 0 to 10193
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Row ID          10194 non-null  int64
 1   Order ID        10194 non-null  object
```

```
 1   Order ID        10194 non-null  object
 2   Order Date      10194 non-null  datetime64[ns]
 3   Ship Date       10194 non-null  datetime64[ns]
 4   Ship Mode       10194 non-null  object
 5   Customer ID     10194 non-null  object
 6   Customer Name   10194 non-null  object
 7   Segment         10194 non-null  object
 8   Country/Region  10194 non-null  object
 9   City            10194 non-null  object
 10  State/Province  10194 non-null  object
 11  Postal Code     10194 non-null  object
 12  Region          10194 non-null  object
 13  Product ID      10194 non-null  object
 14  Category        10194 non-null  object
 15  Sub-Category    10194 non-null  object
 16  Product Name    10194 non-null  object
 17  Sales           10194 non-null  float64
 18  Quantity        10194 non-null  int64
 19  Discount        10194 non-null  float64
 20  Profit          10194 non-null  float64
dtypes: datetime64[ns](2), float64(3), int64(2), object(14)
memory usage: 1.6+ MB

BUSINESS PERFORMANCE SUMMARY
Total Sales: 2326534.35
Total Profit: 292296.81
Total Unique Orders: 5111

STATISTICAL SUMMARY
```

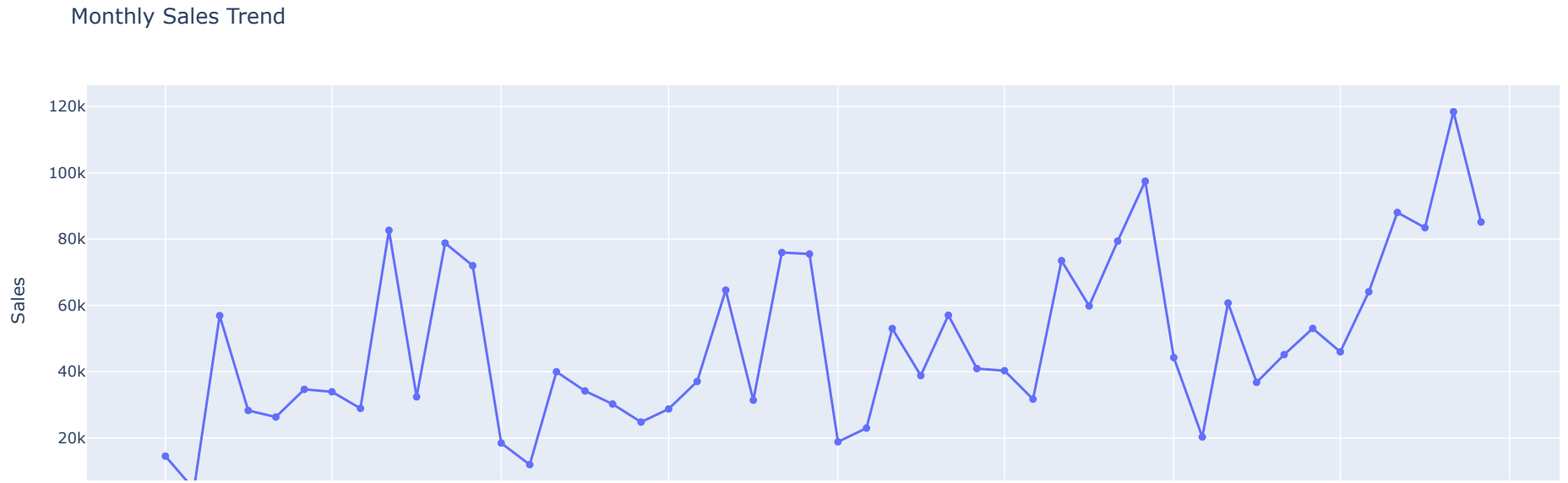
	Sales	Profit	Quantity	Discount
count	10194.000000	10194.000000	10194.000000	10194.000000
mean	228.225854	28.673417	3.791838	0.155385
std	619.906839	232.465115	2.228317	0.206249
min	0.444000	-6599.978000	1.000000	0.000000
25%	17.220000	1.760800	2.000000	0.000000
50%	53.910000	8.690000	3.000000	0.200000
75%	209.500000	29.297925	5.000000	0.200000
max	22638.480000	8399.976000	14.000000	0.800000

SALES BY CATEGORY

Category	Sales
Furniture	754747.7613
Office Supplies	731893.3140
Technology	839893.2790

dtype: float64







Profit by Region



Sales vs Profit

